

PROJECT CHARTER

Project: Financial & Operational Analysis of U.S. Hospitals Using Provider Cost Report Data (2011–2022)

Date: November 3, 2025

Sponsor: Prof. Manjari Maheshwari

Project Manager / Team Leader & Data Integration Lead: Manjunath Muthineni (0878795)

Team: Abhishekh Choudhary – Data Cleaning Lead, Rajesh Thota – Documentation Lead, Krishna Chaitanya Venuturumilli – Analytics Lead, Neha – Finance SME / Analyst

Delivery Mode: GitHub Repository (containing CSV datasets, documentation, dashboards, and reports)

1. Project Purpose & Business Case

Analyze CMS Provider Cost Report data (2011–2022) to generate insights on hospital cost efficiency, resource utilization, and operational performance. Supports data-driven decision-making and policy evaluation. All datasets and KPI metrics will be stored and managed as CSV files within the GitHub repository for transparency and reproducibility.

2. Objectives & Success Criteria

Objectives:

- Clean & harmonize 2011–2022 CMS data into analytics-ready CSV files.
- Develop Tableau dashboards using CSV-based KPI metrics.
- Deliver 8+ insights and 3 strategic recommendations.

Success Criteria:

- Dataset completeness $\geq 90\%$
- Dashboard load time < 4 sec
- Stakeholder satisfaction $\geq 4/5$

3. Scope

IN SCOPE: Data collection, cleaning, transformation, CSV file management, dashboard creation, and insight reporting.

OUT OF SCOPE: Predictive modeling, PHI data, and confidential proprietary datasets.

4. Deliverables (CSV-Based GitHub Repository)

Category	Deliverable	Format / Location	Purpose / Description
Documentation	Project Charter, Business Requirements, Meeting Minutes (Parts 1–4)	/docs/ (PDF/DOCX)	Define project scope, business goals, and record progress
Data Preparation	Data Collection & Cleaning Scripts	/data_collection/, /data_processing/ (Python)	Collect, clean, and transform CMS CSV datasets
Data Assessment	Data Quality & Assessment Report	/data_assessment/assessment_report.csv and .md summary	Quantifies missing values, duplicates, and consistency issues
Processed Datasets	Cleaned Dataset	/datasets/hospital_costs_cleaned.csv	Final unified dataset (2011–2022) used for analytics
Analytics & Metrics	KPI Metrics Catalog	/analytics/kpi_metrics.csv	Contains calculated KPIs like cost per bed, margin per discharge, efficiency ratios
Visualization	Tableau Dashboard File	/visualizations/final_dashboard.twbx	Interactive visualization built from CSV data
Reporting & Presentation	Final Report and Presentation Slides	/reports/final_report.pdf, /presentation/final_slides.pptx	Communicate key findings and recommendations
Handover & Support	README, Runbook, requirements.txt, .gitignore	Root repository	Ensure reproducibility and environment setup
Repository Link	GitHub Repository	To be added upon submission	Central hub for all assets and deliverables

5. Cost Baseline (Role-Based, Junior Team – 4 Months)

Role / Resource Type	Estimated Hours	Rate (CAD/hr)	Estimated Cost (CAD)
Manjunath Muthineni – Team Leader & Data Integration Lead	70	30	2,100
Abhishekh Choudhary – Data Cleaning Lead	110	25	2,750
Rajesh Thota – Documentation Lead	70	22	1,540
Krishna Chaitanya Venuturumilli – Analytics Lead	90	25	2,250
Neha Oberoi – Finance SME / Analyst	60	23	1,380
TOTAL	400 hrs	-	≈ 10,020 CAD

6. Timeline (4 Months)

Month 1: Setup & Data Ingestion → Raw data ready

Month 2: Data Cleaning → 90% data quality achieved

Month 3: Metric Design & Benchmarking → KPI metrics (CSV) ready

Month 4: Dashboards & Reporting → Final insights delivered

7. Stakeholders

Role	Name	Responsibility
Sponsor	Prof. Manjari Maheshwari	Project oversight, milestone approval, grading
Project Manager / Team Leader & Data Integration Lead	Manjunath Muthineni (0878795)	Leads project, integrates datasets, manages timelines and risks
Data Cleaning Lead	Abhishekh Choudhary	Cleans and harmonizes CMS data
Documentation Lead	Rajesh Thota	Maintains project documentation and meeting records
Analytics Lead	Krishna Chaitanya Venuturumilli	Designs KPIs, creates visualizations and Tableau dashboards
Finance SME / Analyst	Neha Oberoi	Validates financial metrics and supports insight interpretation

8. Risks & Ownership

Risk	Impact / Probability	Mitigation	Owner
Data issues / schema inconsistency	High / Medium	Validate early; maintain consistent CSV schema	Data Cleaning Lead
Technical learning curve	Medium / High	Conduct tutorials; peer review scripts	All Team
Missed deadlines	Medium / High	Use buffers; weekly progress check	Project Manager
Dashboard performance	Medium / Medium	Optimize data extracts and filters	Analytics Lead
Misinterpretation of KPIs	Medium / Low	Cross-validation with Finance SME	Finance SME / Analyst

9. Governance & Approval

- Weekly working meetings; weekly sponsor reviews.
- Change >10% scope requires sponsor approval.
- Quality gates: Data ready → KPIs verified → Dashboard published → Final sign-off.

10. Constraints & Dependencies

Dependent on CMS data availability, GitHub access, faculty feedback, and Tableau licensing.

APPROVALS:

Sponsor: _____ Date: _____

Project Manager: _____ Date: _____

(Prepared by: Neha – November 3, 2025)